

User Guide

User Guide

Enterprise-Grade Azure End-to-End Data Engineering Pipeline

Medallion Architecture using Azure Data Factory, Azure Databricks, Azure Synapse Analytics & Power BI

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Table of Contents

1. Introduction
 2. Project Overview
 3. System Architecture
 4. Prerequisites
 5. Azure Resources
 6. Project Structure
 7. Dataset Information
 8. Pipeline Workflow
 9. Running the Project
 10. Monitoring the Pipeline
 11. Power BI Dashboard
 12. Troubleshooting
 13. Best Practices
 14. Future Enhancements
 15. Contact Information
-

1. Introduction

This User Guide explains how to set up, execute, monitor, and understand the **Enterprise Azure End-to-End Data Engineering Pipeline**.

The solution follows the **Medallion Architecture (Bronze → Silver → Gold)** and demonstrates how enterprise data engineering pipelines are built using Microsoft Azure.

2. Project Overview

The pipeline performs the following operations:

CSV Files

↓

Azure Data Lake Storage Gen2

↓

Bronze Layer

↓

Azure Databricks

↓

Silver Layer

↓

Azure Databricks

↓

Gold Layer

↓

Azure Synapse Analytics

↓

Power BI Dashboard

3. System Architecture

The project consists of the following Azure services:

Service	Purpose
Azure Data Lake Storage Gen2	Data Storage
Azure Databricks	Data Processing
Azure Data Factory	Workflow Automation
Azure Synapse Analytics	SQL Analytics
Power BI	Reporting

4. Prerequisites

Before using the project, ensure the following are available:

- Microsoft Azure Subscription
- Azure Data Factory
- Azure Databricks Workspace
- Azure Data Lake Storage Gen2
- Azure Synapse Analytics
- Power BI Desktop
- Git
- GitHub
- Python 3.x
- Dataset

5. Azure Resources

The project requires the following Azure resources:

Resource Group
Azure Storage Account

Azure Databricks

Azure Data Factory

Azure Synapse Analytics

Power BI Desktop

6. Project Structure

```
azure-end-to-end-data-engineering-pipeline/
```

```
|  
├─ architecture/  
├─ notebooks/  
├─ adf/  
├─ powerbi/  
├─ screenshots/  
├─ docs/  
└─ README.md
```

7. Dataset Information

Source Dataset:

Sales Dataset

Files:

- Customers.csv
- Products.csv
- _ProductCategories.csv
- _ProductSubcategories.csv
- Returns.csv
- _Calendar.csv
- _Territories.csv
- _Sales_2015.csv
- _Sales_2016.csv

- `_Sales_2017.csv`
-

8. Pipeline Workflow

Step 1

Upload CSV files to:

Bronze Container

Step 2

Execute Azure Data Factory Pipeline

Pipeline Name:

AZ_ETL_Pipeline

Pipeline Activities:

Bronze_to_Silver

↓

Silver_to_Gold

↓

Gold_to_Synapse

Step 3

Bronze Layer

Stores raw CSV files.

No transformations are applied.

Step 4

Silver Layer

The Databricks notebook performs:

- Remove duplicate records
 - Handle missing values
 - Standardize schema
 - Convert data types
 - Validate data
 - Write Delta tables
-

Step 5

Gold Layer

Creates analytical datasets:

Dimension Tables:

- DimCustomer
- DimProduct
- DimCalendar
- DimTerritory

Fact Table:

- FactSales
-

Step 6

Azure Synapse

Publishes SQL Views:

- vw_DimCustomer
- vw_DimProduct
- vw_DimCalendar

- vw_DimTerritory
- vw_FactSales

Step 7

Power BI

Refresh the report.

Open:

Dashboard.pbix

View business insights.

9. Running the Project

Manual Execution

1. Open Azure Data Factory.
2. Select **AZ_ETL_Pipeline**.
3. Click **Debug** or **Trigger Now**.
4. Monitor pipeline execution.
5. Verify Synapse views.
6. Refresh Power BI.

Scheduled Execution

The pipeline can be configured using:

Azure Data Factory Schedule Trigger.

Execution Flow:

Schedule Trigger

↓

ADF Pipeline

↓

Bronze_to_Silver

↓

Silver_to_Gold

↓

Gold_to_Synapse

10. Monitoring

Pipeline execution can be monitored in:

Azure Data Factory

↓

Monitor

↓

Pipeline Runs

Check:

- Status
- Start Time
- Duration
- Failed Activities
- Error Messages

11. Dashboard Overview

The Power BI dashboard includes:

- Total Orders
 - Total Customers
 - Total Products
 - Total Territories
 - Sales Trend
 - Sales by Country
 - Product Category Analysis
 - Customer Gender Distribution
 - Top Products
 - Interactive Filters
-

12. Troubleshooting

Pipeline Failure

Verify:

- Linked Services
 - Storage Credentials
 - Databricks Cluster
 - Synapse Connection
-

Databricks Error

Verify:

- Cluster is Running
 - Notebook Path
 - Storage Access
 - Delta Tables
-

Synapse Error

Verify:

- SQL Views

- Workspace
 - Authentication
-

Power BI

Verify:

- Data Source
 - Refresh
 - Synapse Connection
-

13. Best Practices

- Use Delta Lake for storage.
 - Keep Bronze data immutable.
 - Validate data before loading into Silver.
 - Store only business-ready datasets in Gold.
 - Parameterize notebooks.
 - Use Azure Key Vault for secrets.
 - Monitor pipeline runs regularly.
-

14. Future Enhancements

- Event-based Trigger
 - Incremental Loading
 - Delta MERGE
 - Azure Key Vault
 - CI/CD with GitHub Actions
 - Azure DevOps
 - Infrastructure as Code
 - Logging Framework
 - Monitoring Alerts
 - Data Quality Framework
-

15. Contact

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Document Version

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1.0	July 2026	Initial Release

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